
Bishoksan Kafle, PhD

STEM Leader | Research Fellow & Scientist | Software Engineer | National Security Technologist | Co-founder [NeDS](#)

📍 Canberra ACT | Australian Citizen | Baseline Security Clearance (Exp: Aug. 2037)
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Professional Summary

- ★ Accomplished Researcher and Tech Professional with a PhD in Computer Science and 10+ years of experience in Automated Reasoning, AI, Program Analysis & Verification across Academia, Government, Non-profit, and Industry.
- ★ Global Leader with a strong track record of leading cross-disciplinary teams, managing high-impact research projects, and developing innovative solutions to address national security challenges and defend Australia's national interests.
- ★ Skilled in stakeholder engagement, collaboration, and communication, with a commitment to continuous professional development and applying research to inform public policy and address national security challenges.
- ★ Passionate about applying scientific rigour to public policy, strategic decisions, and national security. Committed to advancing technology and leadership in the public sector, while continuously pursuing professional development to stay ahead of emerging trends and best practices to protect Australia's interests.

Key achievements

- ★ **Communication Skills** | Delivered over 40 scientific presentations globally, including a keynote talk at NASA Ames Research Center, effectively translating complex research into actionable insights for academic, industry, and government audiences.
- ★ **Research Leadership and International Recognition** | Lead author of 25+ peer-reviewed [publications](#), including two books, advancing software reliability and AI, with 250+ citations from leading institutions such as Oxford, Stanford, Microsoft, and Google; awarded Best Paper at PEPM 2015 for impactful work in semantic program transformation. Led and shaped global research through chairing international conferences ([LOPSTR](#), [HCVS](#)), serving on 12+ program committees, acting as accessor for the ARC, and as Steering Committee Member of LOPSTR series -- contributing to innovative solutions in national security and informing public policy through applied research.
- ★ **Automation for Operational Efficiency and National Security** | Led the design and implementation of an automated tool that significantly improved operational efficiency within critical national programs at the Department of Home Affairs. The solution reduced manual workload and errors, saved thousands of staff hours, and enhanced data-driven decision-making—contributing directly to program effectiveness and supporting national security objectives.

Education & Qualifications

PhD in Computer Science (*Danish Govt. scholarship*) - Roskilde University, Denmark (2012 - 2016)

- Thesis: Components for automatic Horn clause verification (Adv: Prof. John P. Gallagher)

European Master's in Computational Logic (*Erasmus Mundus Scholarship*) - Dresden University of Technology, Germany, Free University of Bozen-Bolzano, Italy, and New University of Lisbon, Portugal (2010 - 2012)

- Thesis: Worst-case execution time of programs (Adv: Prof. Pedro Barahona & Franck Cassez)

Bachelor's in Computer Science (*Nepalese Govt. scholarship*) - Central University of Las Villas, Cuba (2004-2009)

Research Leadership, Collaboration & Communication

- **Research leadership & excellence** | 25+ peer-reviewed publications (incl. 2 books) and 10+ advanced research tools in AI, program analysis, verification, and rule-based reasoning. International leader in AI and automated reasoning with roles on 15+ program committees, 2+ program chairs, steering committee member of a conference series and regular reviewer for 5+ top journals.

- **Strategic Project Delivery & Impact** | Led and successfully delivered 6+ large-scale international projects funded by NASA, the EU, and govt. of Australia, Spain, and Italy, showcasing expertise in managing complex, multi-stakeholder and high-impact projects. Skilled in translating technical research into strategic policy insights for senior executives.
- **Community Engagement & Contribution** | ARC assessor, honorary researcher at the University of Melbourne, and co-founder of Nepal Development Society — shaping national research funding and priorities, mentoring emerging researchers and supporting development through research, collaboration, and mentorship.
- **Sharing knowledge and Mentorship** | Delivered 40+ scientific talks, supervised 20+ graduate students -- driving knowledge transfer, fostering innovation and nurturing the next generation of STEM leaders.
- **Collaboration & Facilitation** | Chaired conferences, facilitated strategic planning, and led delivery of impactful projects and training programs driving research impact and enhanced learning outcomes.

Technologies and skills

Programming Languages	Years of experience	Self-Rating	Middleware Technologies	Years of experience	Self-Rating
Procedural (Java, C, C++, Python, Pascal)	12+	★★★★	IBM MQ, IIB, IBM ACE, ESQL, Broker Flows	4+	★★★★
Functional (Ocaml)	4+	★★★★			
Logical (Prolog)	10+	★★★★★			
Web Technologies			Constraint Solving Techniques		
JEE, JSF, JSP, Hibernate, JavaScript, CSS, HTML, DBs	7+	★★★★	SAT/SMT Solving, Constraint Solving and Optimisation	8+	★★★★★
Software Verification Technologies			Document Processing		
Abstract Interpretation, Model Checking, Semantic Program Transformation, Static Analysis, Automated Reasoning	11+	★★★★★	Microsoft Packages, Latex	15+	★★★★★
Conference Management			Database		
EasyChair, HotCRP	6+	★★★★★	SQL, MySQL, Oracle, IBM DB2, SQL Server	10+	★★★★
Source Control			CI/CD		
SVN, Git, Gitlab	8	★★★★★	ADO, Gitlab	4+	★★★★★
IDE			Data Analysis		
NetBeans, Eclipse, VisualStudio, Jupyter Notebook	7+	★★★★	Python, SPSS	5+	★★★★

Relevant Professional experience

Development and Programming Officer

2022 - Present

Department of Home Affairs

ACT, Australia

Contributing to critical national programs and initiative through data science, automation, software systems, and cross-functional collaboration in a high-stakes environment.

- ★ **Cybersecurity leadership** | Led our section's integration into departmental cybersecurity hardening efforts, aligning with the **Government's Essential 8**, improving risk posture and compliance across sensitive systems.
- ★ **Strategic Impact** | Delivered automated solution that significantly reduced manual workloads, improved operational efficiency and enhanced data-driven decision-making—contributing directly to program effectiveness and supporting national security objectives.

- ★ **Intelligence & Data Management** | Oversee data management for Cargo and PNRGOV systems, ensuring quality, compliance, and security. Cargo supports \$26B+ in annual revenue; PNRGOV enables real-time risk assessment and national security through advance passenger data.
- ★ **Capability building** | Currently serving/served as a recruitment panel member for APS5 and APS6 positions, contributing to strategic hiring decisions and talent acquisition.

Co-Founder and Executive Committee Member

2013 - Present

Nepal Development Society (NeDS)

Providing strategic and technical leadership to advance sustainable development, public health, and environmental initiatives in Nepal—applying research to real-world impact.

- ★ **Strategic Leadership** | Helped define the organisation's vision and research agenda, contributing to long-term planning and measurable impact in governance, policy, and community resilience.
- ★ **Systemic positive impact** | Empowered local communities and improved governance by fostering collaboration between government, civil society, and the private sector, informing more inclusive policies and change in Nepal.
- ★ **Program Design & Implementation** | Lead public health and community development initiatives—including diabetes and hypertension prevention, sustainable urban farming, and rural health awareness programs—applying research insights to design and guide evidence-based interventions

Research Fellow in Computer Science

2016 - 2022

The University of Melbourne, and IMDEA Software Institute

Australia & Spain

Led and contributed to impactful research efforts at the intersection of software reliability, automated reasoning and program analysis aligned with Defence science objectives.

- ★ **Tool Development & Innovation** | Developed advanced research tools and participated in tools competition, significantly advancing the fields of software reliability, program analysis and verification globally.
- ★ **Securing funds and strengthening quality** | Secured and led \$2M+ in competitive research funding, expanding research capabilities, and enhancing the quality and impact of academic research.
- ★ **Research Impact** | Co-authored 8+ peer-reviewed papers on program analysis, verification and decision procedures focused on safety- and mission-critical systems and presented at conferences.

Visiting Research Scientist

Mar – Jun 2015

NASA Ames Research Center, Intelligent system Division

California, USA

- ★ **Learning and Development** | Gained experience in NASA avionics systems and conducted resource analysis research under renowned guidance, resulting (later) in two published articles and the development of a research tool.
- ★ **Collaboration** | Presented findings at NASA, fostering valuable collaborations with researchers. These contributions advanced the understanding of avionics software and strengthened connections within the research community.

Software Engineer

2009 - 2011

F1 Soft International

Nepal

Provided software development services and played a key role in developing [e-Sewa](#), a widely used payment gateway in Nepal, leveraging business insight and cutting-edge technologies.

- ★ **Systems Improvement** | Successfully enhanced over 5 applications by integrating SMS operations and developed more than 11 mobile applications, including a mobile version of e-Sewa.

Certifications & Training, Presentations and Publications

Certifications & Training:

- *Artificial Intelligence in Government: Fundamentals*, Australian Government (2024)
- *Building and Learning High Performing Teams*, APS Academy (2024)
- *Building Relationships and Engagement*, APS Academy (2024)
- *Effective Management Conversations*, APS Academy (2025)
- *Crafting Quality New Proposals – Practitioner*, APS Academy (2023)
- *Planning and Managing Change – APS Academy* (2024)
- *Workplace Communication in Practice*, APS Academy (2023)

Selected Presentations (of a total 40+):

- *Static Resource Analysis with Application to Avionics Systems* to 30+ research scientists and business leaders at NASA Ames Research Center, CA, US (2015)
- *Transformation-Enabled Precondition Inference* to 137+ research scholars for ICLP Conference, online (2021)
- *A Benders Decomposition Approach to Deciding Modular Linear Integer Arithmetic* to 149+ research scholars for SAT Conference, Melbourne Australia (2017)
- *RAHFT: A Tool for Verifying Horn Clauses Using Abstract Interpretation and Finite Tree Automata* to 190+ research scholars for CAV Conference, Toronto, Canada (2016)

Selected [Publications](#) (of total 25+):

- **[Journal]** Bishoksan Kafle, Graeme Gange, Peter J. Stuckey, Peter Schachte & Harald Sondergaard: *A Lightweight Approach to Nontermination Inference Using Constrained Horn Clauses*. Software and System Modeling (2024).
- **[Book]** Robert Glück & Bishoksan Kafle: *Logic-Based Program Synthesis and Transformation - 33rd International Symposium*, LOPSTR. LNCS 14330, Springer 2023, ISBN 978-3-031-45783-8.
- **[Journal]** Bishoksan Kafle, John P. Gallagher, Graeme Gange, Peter Schachte, Harald Sondergaard & Peter J. Stuckey: *An iterative approach to precondition inference using constrained Horn clauses*, TPLP (2018).
- **[Journal]** Bishoksan Kafle & John P. Gallagher: *Constraint Specialisation in Horn Clause Verification*, Science of Computer Programming (2017).